

Machine Learning and Agentic AI for Macroeconomic Research

Lec. 10: Case Studies — Agentic AI in Economic Research

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Spring 2026

Lecture Agenda

1. Why case studies after Lec. 9
2. Three research patterns
3. Case Study 1: paper-review skill
4. Case Study 2: MEPS uninsurance
5. Case Study 3: ES fellows dataset
6. Case Study 4: AI as research collaborator
7. Case Study 5: inside the sprint (process walkthrough)
8. Synthesis across the cases
9. Exercise A: guided MEPS lab
10. Exercise B: fellows assignment
11. Appendix: prompts, schema, and references

Teaching claim: agents help economists in three concrete ways — reusable quality-control workflows, empirical data pipelines, and provenance-aware public-web dataset construction — and one meta way: as a full-lifecycle collaborator whose workflow is itself a studied object (Case 5).

Framing

From Lec. 9 concepts to complete research workflows

Where We Are in the Course

1–6: models + AI/ML tools

7–8: LLMs + RAG

9: agentic AI workflow

10: case studies

- Lec. 9 gave the workflow: specification, structure, validation, and human oversight
- Lec. 10 shows how that workflow changes shape across different research jobs
- The core question is no longer “what is a skill?” It is “what workflow should I build for this task?”

Lab Readiness Check

Before today's exercises, verify all five:

1. **Claude Code installed** — `claude -version` returns a version number
2. **Subscription active** — Pro or Max at `claude.ai`
3. **Git initialized in your project** — `git status` runs without error
4. **Working Python or R environment** — `which python` or `which R` returns a path
5. **CLAUDE.md created** — one file with your project name, language, and goal

If any box is unchecked, open Appendix A now. The exercises in this lecture require all five. If setup takes more than 10 minutes, pair with a neighbor and share one terminal.

Why Case Studies After Lec. 9

- Knowing the vocabulary of agentic AI does not tell you how to design a research workflow
- The difficult step is decomposition: where to split the task, where to validate, and what stays human
- Different tasks need different harnesses: paper review is not the same as MEPS harmonization, and neither is the same as public-web metadata collection
- Today is about design judgment, not product loyalty

Question for every case: what is the artifact, what is the failure mode, and what must the economist still verify?

Three Research Patterns

Pattern	Representative task	Main risk if you use the wrong workflow
Reusable skill	Referee-style paper review	Generic comments, hallucinated quotes, missing quality gate
Empirical pipeline	MEPS harmonization and figures	Wrong crosswalk, dropped survey weights, silent validation failure
Public-web dataset	Econometric Society fellows metadata	Broken source discipline, missing provenance, hidden missingness

All three use agents, but the harness is different.

What Remains Human Across All Three

You still own

- the research question
- the coding rules
- the validation target
- the interpretation
- the decision to trust or reject the output

The agent accelerates

- repetitive execution
- file handling
- promptable decomposition
- re-runs after fixes
- draft diagnostics and plots

Lec. 9 still governs this lecture: specification and validation are not optional just because the execution is faster.

Case Study 1

A reusable quality-control workflow for paper review

Case Study 1: Why This Becomes a Skill

Research job: produce a structured referee-style review for a quantitative economics paper with quote-level evidence and quality gates.

- You may run this workflow many times across papers, coauthors, and seminars
- Quality matters more than speed because a wrong criticism is costly
- The task has internal stages with clear dependencies
- That makes it a good candidate for a reusable skill rather than a one-off prompt

The Naive Baseline Fails Fast

Tempting prompt: “Review this paper.”

What usually comes back:

- generic praise and generic weaknesses
- no verbatim anchor to the paper
- no proof-checking discipline
- no calibration to quantitative macro standards
- no filter removing low-confidence comments

The failure is not just model quality. It is the absence of structure, specialization, and a final editorial gate.

What a High-Quality Output Should Look Like

Shortened review template

`## Overall Feedback`

- `1. Estimand is not fully pinned down in Section 2.3`
- `2. Proof of Theorem 2 needs one missing justification`
- `3. Numerical section needs a reproducibility note`

`## Detailed Comments`

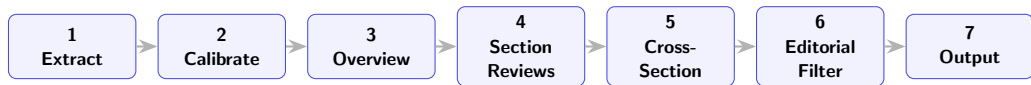
`### 1. Proof step in Theorem 2`

`Quote: "By dominated convergence ..."`

`Feedback: This step needs an explicit domination argument.`

Target properties: issue prioritization, exact anchoring, specific fixes, and a recommendation you can defend.

The 7-Phase Pipeline



- Parallelism happens inside phases, not across them
- The output of one phase becomes the acceptance criteria for the next
- This is a quality-control pipeline, not just a prompt chain

Why Decomposition Matters Here

Context

- long paper
- review rules
- output template

One call is too coarse.

Specialization

- proof review
- methodology review
- literature fit

One prompt does not do all well.

Quality

- generic comments
- invented quotes
- duplicated issues

You need a final filter.

Economics translation: this is like assigning specialized RA tasks, then forcing everything through one senior review before it goes out.

The Highest-Value Guardrails

1. **Confidence gate.** Do not assert an error unless the model can derive it step by step.
2. **Quote verification.** Comments that cite the paper must be anchored to real text.
3. **Editorial filter.** Remove generic advice, duplicates, and stylistic fluff.
4. **Humanization.** The final report should read like a serious economics referee report, not model boilerplate.

The lesson is broader than paper review: when the stakes are high, validation must be a named stage in the workflow.

Worked Example: Why the MPE Paper Is a Good Stress Test

Demo paper: *Markov Equilibria with Quasi-Hyperbolic Discounting*

- it mixes proofs, numerical work, and dynamic economic interpretation
- it forces the workflow to separate theorem checking from broader conceptual review
- it is hard enough that a one-shot prompt should fail

This is why case studies matter: you only learn the workflow once you see where the naive baseline actually breaks.

What the Economist Still Owns in Case Study 1

Agent side

- parse sections
- fan out specialized prompts
- draft comments
- enforce formatting

Economist side

- decide what counts as a serious objection
- reject false positives
- judge novelty and importance
- decide what feedback is worth sending

Skill logic: you are not outsourcing judgment. You are outsourcing disciplined first-pass execution.

When Should a Workflow Become a Skill?

Build a skill when ...	Stay ad hoc when ...
you will repeat the workflow many times	the task is one-off
the workflow has named validation stages	you are still exploring what the workflow should be
multiple users need consistent output	only one user needs the result
quality failures are expensive	output format is flexible

Case 1 verdict: this is a skill because repeatability and quality control are the point.

Case Study 2

A one-session empirical data pipeline with strong validation

Case Study 2: The Macro Question

Question: Who is uninsured in the United States, 2000–2023, and how did the ACA change the distribution?

- health insurance is a major incomplete-market margin
- the ACA is a large social-insurance intervention
- the research value comes from who gained coverage, not just from the overall average
- this is the most economics-facing case in the lecture

The Naive Baseline: 24 Files, 24 Headaches

- download 24 MEPS full-year files by hand
- discover that variable names are year-suffixed
- figure out which weight variable applies in each year
- merge everything and hope the crosswalk is right

Where this breaks:

1. silent crosswalk mistakes
2. dropped survey weights
3. inconsistent file formats
4. wrong figure even though the code runs

Why This Stays Ad Hoc Instead of Becoming a Skill

Skill-like features

- multi-step pipeline
- explicit validation
- repeated checks

Why still ad hoc

- single research question
- only three prompts
- interactive refinement of plots
- workflow is useful but not yet a reusable product

Case 2 verdict: this is a structured session, not a reusable skill.

The 3-Prompt Architecture

Prompt A: download, parse, and harmonize MEPS files

Prompt B: compute weighted uninsurance rates by group

Prompt C: generate publication-quality figures and annotations

- Phase A builds infrastructure
- Phase B turns infrastructure into statistics
- Phase C turns statistics into economic evidence

Phase A: What the Agent Actually Does

- discovers the AHRQ download pattern
- retries around 403 or parser failures
- builds a year-specific crosswalk
- standardizes the merged output into one panel

This is the boring but decisive part. If the infrastructure is wrong, every figure downstream is wrong in a disciplined-looking way.

Schema Harmonization Is the Hard Part

Concept	2001	2014	2022
Insurance coverage	INSCOV01	INSCOV14	INSCOV22
Person weight	PERWT01F	PERWT14F	PERWT22F
Age	year-specific	AGELAST	AGELAST

- humans get tired and miss one mapping
- agents are good at repetitive lookups if you force them to save the crosswalk
- validation is still essential because a bad crosswalk can look plausible

Weights Are Not Optional

$$\text{Uninsurance rate} = \frac{\sum_i w_i \cdot \mathbf{1}[\text{INSCOV}_i = 3]}{\sum_i w_i}$$

- MEPS is a complex survey, not a simple random sample
- unweighted rates distort the national picture
- the prompt must state the weighting rule, and you must still inspect the code

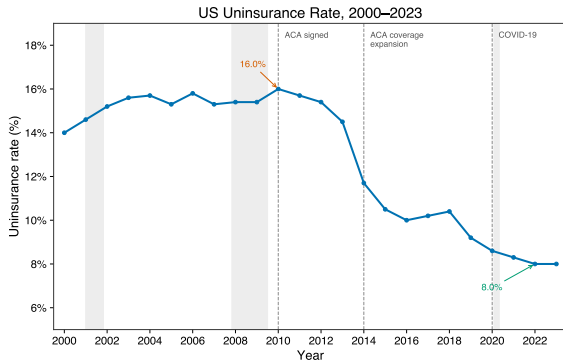
Case 2 lesson: a fast agent does not relieve you of basic survey-design discipline.

Validation: Trust But Verify

Year	External benchmark	What your MEPS output should show
2010	CPS ASEC about 16%	high pre-ACA uninsurance
2013	CPS ASEC 13.4%	near the pre-ACA peak
2016	CPS ASEC 8.8%	clear post-2014 decline
2019	CPS ASEC about 8%	stable late-2010s level
2022	CPS ASEC about 8%	still low relative to 2000s

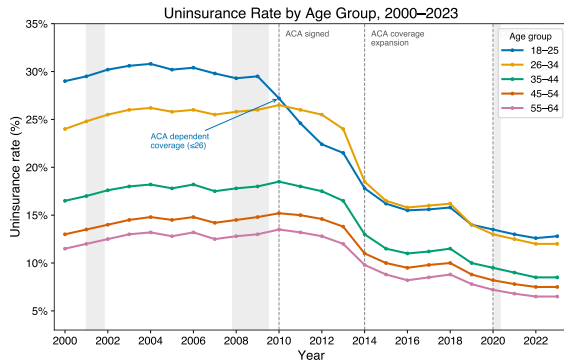
The economist's job: know enough about the data that a 5% or 25% result for 2013 looks obviously wrong.

Figure 1: Overall Uninsurance Rate, 2000–2023



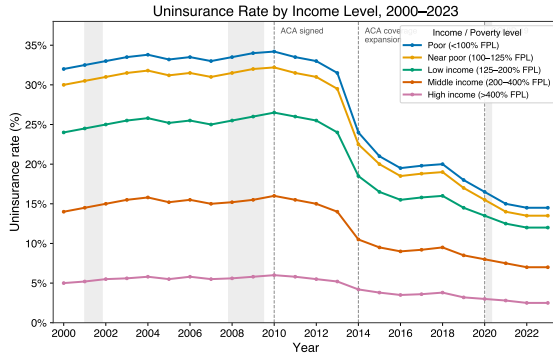
Headline: high through the 2000s, then a sharp decline after the 2014 ACA coverage expansion.

Figure 2: By Age Group



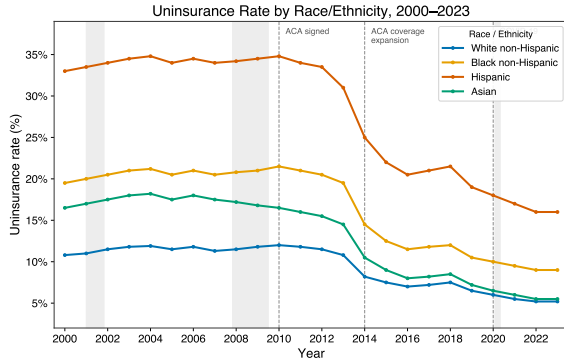
Economic reading: the life-cycle profile is steep, and the post-ACA drop is not uniform across ages.

Figure 3: By Poverty Category



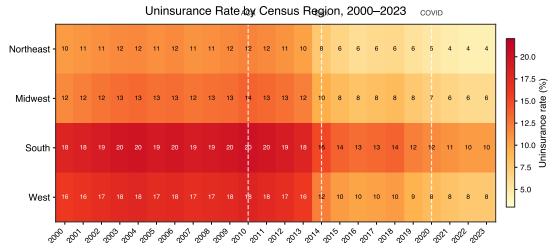
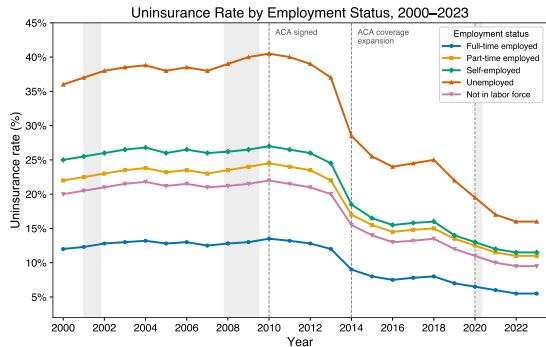
Economic reading: the largest levels and some of the largest gains are concentrated lower in the income distribution.

Figure 4: By Race and Ethnicity



Economic reading: the ACA compresses disparities, but it does not erase them.

Figures 5 and 6: Employment and Region



Takeaway: one pipeline can generate multiple cuts of the same policy story once the harmonized panel exists.

What Claude Did vs. What Stayed Human

Agent

- downloaded files
- handled parsing obstacles
- built the crosswalk
- computed weighted statistics
- rendered figures

Economist

- chose the question
- specified the weights
- chose the subgroup cuts
- validated against benchmarks
- interpreted the policy story

Exercise A: Guided MEPS Lab

- keep the in-class lab short and structured
- students run the same three-prompt arc from the lecture
- success criteria: one merged file, weighted group-level CSVs, and at least three validated figures
- the notebook is the operational handout; the slides only carry the logic

Pedagogical point: the lab is about directing the agent and checking the output, not about typing code by hand.

Case Study 3

A provenance-aware public-web dataset construction workflow

Why This Third Case Is Different

- the source material is public and semi-structured
- metadata quality is uneven across people and institutions
- missingness is part of the result, not a nuisance to hide
- the workflow must control sources, provenance, and coding rules

This is not a web-scraping stunt. It is a data-construction problem with explicit source restrictions.

Research Question for the Fellows Dataset

Question: What can we learn about career stage, election timing, and field composition among current Econometric Society fellows using only official and institutional sources?

- it is a useful dataset about the economics profession itself
- it exposes a new research pattern: roster seed + metadata enrichment + coverage reporting
- it forces us to separate current biological age, years since PhD, and age at fellow election
- it forces students to confront missingness, source quality, and coding judgment

Why a Naive Web-Chat Workflow Still Fails Here

1. **Roster control is weak.** You need a canonical seed list and fixed batches.
2. **Source discipline is weak.** You need a rule that excludes Wikipedia and open-web guesswork.
3. **State tracking is weak.** You need persistent files recording what was found and what stayed missing.
4. **Auditability is weak.** You need field-level source URLs, not just narrative answers.

Precise contrast: modern web chat can browse and summarize, but this assignment needs a project harness with files, batching, and provenance rules.

Official Seed Roster as of April 14, 2026

- canonical source: the official Econometric Society current fellows page
- parsed roster size: 882 fellows
- official CV links on the seed page: 64
- usable election years in the current page: 880; two entries parse as 0000
- batching rule: 18 batches of 50 fellows
- seed fields: name, affiliation, election year, profile URL, CV URL

Interpretation:

- the roster is large enough that the workflow must be batched
- CV coverage is sparse enough that missingness will remain visible
- the current roster is not the same thing as a confirmed-living roster

Batch Workflow and Project Structure

Project files

- seed roster CSV
- batch assignment file
- normalized batch CSV
- analysis script
- plots and memo

Workflow

1. parse the official roster
2. assign a batch of 50
3. fill official/institutional metadata
4. run analysis
5. write findings and limitations

This is the public-web analog of 'CLAUDE.md': the project files hold the rules and state across sessions.

Schema and Provenance Rules

Field	Rule
name, affiliation, election_year	seed roster only
bio_url, cv_url	official or institutional only
phd_year, re-search_fields_raw	record source URL for each field
normalized field bucket	map raw text into the lecture taxonomy
coverage_status	say what remains missing
notes	document coding judgment or ambiguity

Credibility requires traceable provenance for each field.

Missingness, Gender, and Living-Status Caution

- **Birth date:** leave missing unless the official source states it
- **PhD year:** use only if the source states the degree year directly
- **Gender:** code only from explicit pronouns or explicit gender language
- **No inference:** not from names, photos, or country
- **Living status:** do not infer from old election year alone; for lecture purposes, use a 90-year cap only as a conservative sensitivity rule

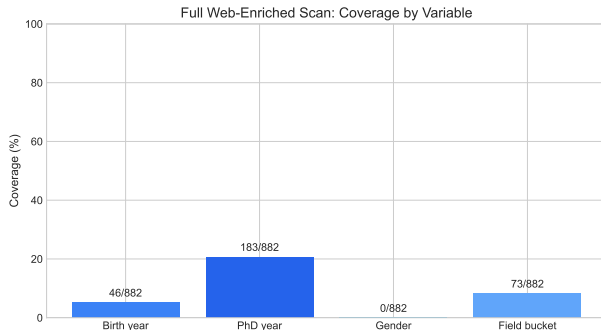
This is deliberate. The assignment is designed to teach credible coding rules, not aggressive completion.

Instructor Full Scan: What the Agentic Workflow Added

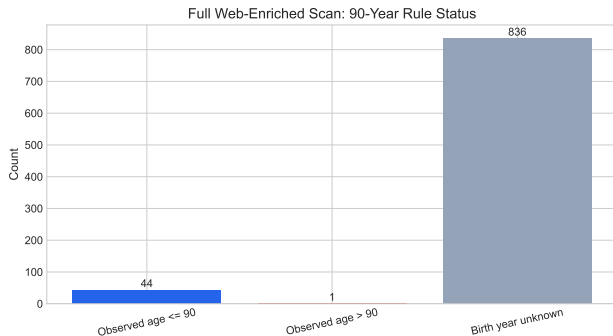
Metric	Official → Web
Local CV docs	62 → 316
Birth year	11 → 46
PhD year	47 → 183
Field bucket	15 → 73

Main implication:

- the agent scaled discovery and parsing well
- PhD-year coverage improved materially
- birth-year coverage stayed thin



The 90-Year Rule Does Not Solve the Main Problem

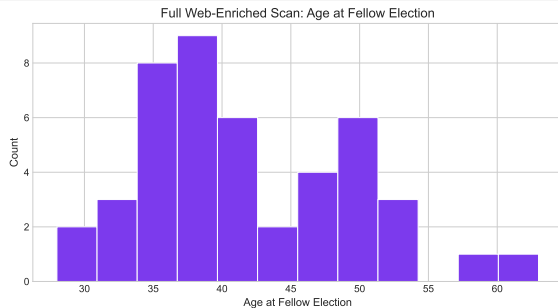
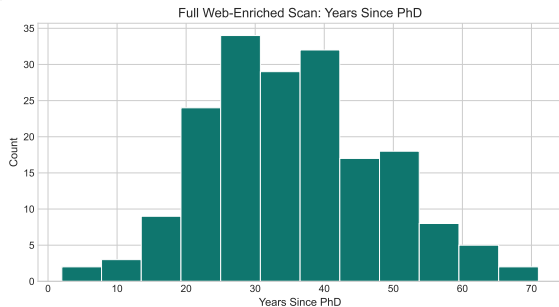


- observed age ≤ 90 : 44 fellows
- observed age > 90 : 1 fellow
- birth year unknown: 836 fellows

Takeaway:

- the problem is missing birth year, not a handful of extreme ages
- do not treat current biological age as the main variable
- use age at election or years since PhD instead

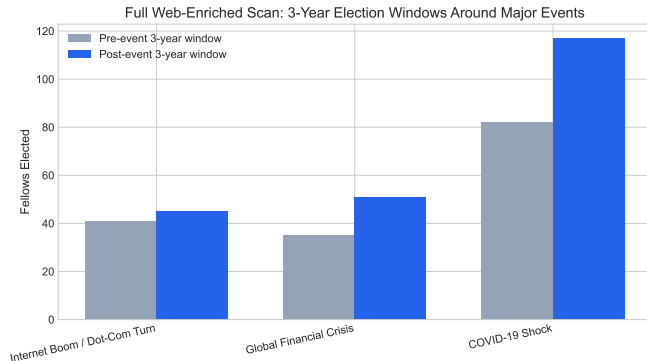
Career Stage: Use Years Since PhD and Age at Election



Main descriptive findings:

- median years since PhD: 35
- median age at fellow election: 40
- median years from PhD to election: 17
- age at election is cleaner than current age because it does not depend on fellows being alive today
- this is exactly why agent-generated datasets still need variable-design judgment

Election Counts Around Major Economics Events



3-year post-event windows

- Internet turn (2000): 41 → 45
- GFC (2008): 35 → 51
- COVID era (2020): 82 → 117

Interpretation:

- there is no obvious post-dot-com collapse
- counts rise after 2008 and surge after 2020
- the 2020–2024 jump likely reflects Society-side policy or catch-up effects
- use this as descriptive timing evidence, not a causal event study

What Agentic AI Actually Did Here

- discovered and cached candidate homepages at scale, then kept the search state on disk
- parsed 527 institutional profile pages and downloaded 316 CV or CV-like documents
- normalized fields, tracked source URLs, and reran the analysis after rule changes
- surfaced a negative result clearly: even a large search pass does not rescue current biological age

What stayed human:

- define source restrictions and coding rules
- separate current age, years since PhD, and age at election
- treat event patterns as descriptive rather than causal
- decide when missingness is too severe for a claim

Exercise B: Full-Roster Fellows Assignment

- start from the official seed roster and take one batch of 50 fellows
- use only official and institutional sources
- produce a normalized CSV, four figures, and a short memo
- keep missing values visible and documented
- use age only if birth-year coverage reaches the lecture threshold; otherwise use years since PhD or age at election
- explain clearly what the agent automated and what judgment you supplied

This case is not about scraping harder. It is about designing a trustworthy public-web research pipeline.

Case Study 4

AI as full-lifecycle research collaborator in empirical macroeconomics

Case 4 — The Research Question

Question: Why has rapid AI adoption not produced uniform occupational collapse, and what role does coordinative labor play in shaping which occupations survive?

- Paper: “Glue Work, Task Bundles, and the AI Labor Market” — Feng (2026, in progress)
- Extends Garicano–Li–Wu (2026) by endogenizing bundle strength via glue labor supply
- **This case covers the full research lifecycle:** theory articulation, empirical pipeline design, adversarial review, and claims discipline — not just one execution stage

The Glue Work Hypothesis

What is glue work?

- Coordinative tasks that integrate components without producing a visible standalone output
- O*NET examples: coordinating work and activities; scheduling; communicating across silos
- Underrewarded: workers capture only $\lambda < 1$ of the social marginal product
- When glue erodes, the bundle weakens — often with a lag before it becomes visible

The 2×2 AI impact matrix

	Low glue	AI	High glue	AI
High AI prod.	Restructure		Full disruption	
Low AI prod.	Stable		Glue erosion	

Core prediction: high-AI + high-glue occupations *restructure* rather than collapse.

The insight was human. The matrix formalization was joint.

Stage 1 — Theory Building with AI

- Human developed the core insight: endogenize bundle strength through glue labor supply
- Claude was asked to:
 - Articulate the 2×2 prediction matrix from verbal intuition
 - Define AI-Easy Glue vs. AI-Hard Glue using specific O*NET item codes
 - Identify the market failure formally ($\lambda < 1$ creates underinvestment equilibrium)
- **AI-Easy Glue** (LLM-susceptible): documenting/recording information, synthesizing knowledge, routine cross-functional reporting
- **AI-Hard Glue** (embodied/relational): building trust, mentoring, navigating ambiguous authority, reading implicit organizational signals
- Prompting mode: **specification-driven** — “List O*NET items 4C1–4C3 that qualify as AI-Hard Glue by the following criteria...” not “tell me what glue work is”

Stage 2 — Translating Theory into Empirical Specification

```
## Empirical Specification Prompt
```

```
You are translating the following theoretical model into a  
testable empirical specification.
```

```
THEORY CLAIMS:
```

- P1: High-glue, high-AI-exposure occupations restructure rather than collapse (employment share positive 2019-2024)
- P2: The reallocation margin operates at junior entry (age < 30), not incumbent exit

```
VARIABLE CONSTRUCTION RULES:
```

- glue_core: mean of O*NET items [4C1.a.2, 4A4.a.1, 4A4.b.1] normalized to [0,1] at SOC 2018 level
- ai_exposure: FRS AIOE score, 2023 vintage
- outcome: CPS monthly employment share (seasonally adj.)

```
FORBIDDEN: do not use occupation-level wage as a glue proxy  
without first checking correlation > 0.6 with glue_core.
```

- Exact item codes prevent hallucinated variable definitions
- A “FORBIDDEN” instruction enforces scope before results appear

Claims Discipline — Setting the Rules Before Seeing Results

```
## Claims Discipline Specification

BEFORE any data is analyzed, record the following:

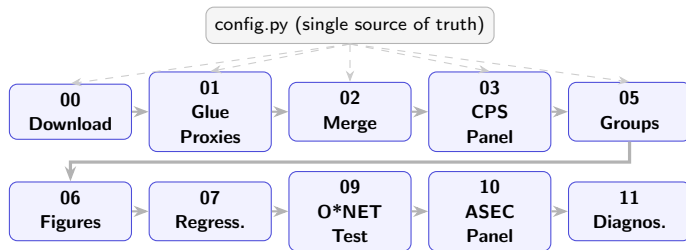
ALLOWED CLAIMS (theory-derived, testable):
- P1: Occupations in top quartile of both glue_core and AIOE
  show positive employment share growth 2019--2024
- P2: Entry share (age < 30) declines faster in
  low-glue / high-AI occupations

FORBIDDEN CLAIMS (data-mining temptations to block):
- Any claim about wage effects not pre-specified here
- Any interaction term not listed in the model section
- Any subgroup finding discovered post-estimation

GATE: results memo must cite which allowed claim each
regression speaks to before interpretation proceeds.
```

This is claims discipline. It must be set before results appear, not reconstructed after. It is the empirical analog of pre-registration.

Stage 3 — The 11-Script Pipeline Architecture



One config.py = single source of truth. 51 unit tests specified before any script runs on real data.

Pipeline Module Detail

Script	Key output	Validation rule	Each
00	onet_raw, frs_aioe, crosswalks (parquet)	row count $\geq$ expected; no null SOC codes	
01	glue_proxies (6 variants, SOC 2018)	all variants in $[0, 1]$; correlation matrix saved	
02	master_soc: glue + AI exposure + crosswalks	merge rate $\geq 95\%$; no duplicate SOC	
03	cps_panel: employment share, entry share	weights sum to civilian pop; seasonal adj. flag set	
05	analysis: 4 AI $\times$ glue groups defined	group counts balanced; no SOC in > 1 group	
06–11	figures, tables, diagnostics	files written; no NaN in reported statistics	

script has explicit inputs (parquet), explicit outputs (parquet), and a pass/fail validation rule. Failure halts the pipeline.

Stage 3 — Variable Construction Details

Glue proxies (6 variants)

- `glue_core`: 3 O*NET items, baseline
- `glue_ext`: 8-item extended set
- `glue_aihard`: AI-Hard items only
- `glue_social`: social/interpersonal subset
- `glue_dominance`: mean glue / mean all activities
- `glue_minus_mgmt`: robustness check

The 6 variants test whether the theory's protection story depends on which aspect of coordination is measured.

AI exposure and outcomes

- AI exposure: FRS AIOE 2023, SOC-linked
- Outcomes: CPS monthly employment share
- Entry margin: age < 30 share within occ-month
- Long-run: ASEC panel 2000–2025
- 4 groups: median splits of AIOE $\times$ glue_core

Core group definitions:

High AI + High Glue	Restructuring
High AI + Low Glue	Disruption
Low AI + High Glue	Insulated
Low AI + Low Glue	Stable

Stage 4 — Adversarial Review

Adversarial prompt: “Find 6 structural weaknesses in this empirical design. For each, state whether it is fatal or manageable, and why.”

Six weaknesses identified:

1. AIOE is cross-sectional — does not capture within-occupation AI adoption dynamics
2. Glue proxies are task-level O*NET ratings, not worker-level observations
3. Selection: high-glue workers may self-select into high-AI firms for unobserved reasons
4. Crosswalk from SOC 2018 to Census codes loses $\sim 12\%$ of occupations (a validation rule catches this)
5. CPS monthly sample sizes for narrow occupations produce large standard errors
6. **The interaction ($\text{AI} \times \text{Glue}$) is negative in the data** — but the theory predicts positive

Item 6 was the decisive finding. It repositioned the entire paper.

When Data Contradicts Theory

Data finding: the estimated interaction (AI exposure $\times$ glue score) is *negative*.

Theory predicted: positive — glue preserves bundles, so high-glue/high-AI should survive.

This is not a bug. It is information.

What the theory predicted

- High-AI + High-Glue $\rightarrow$ restructuring and survival
- Glue preserves value when production tasks are automated
- Expected positive interaction

The reinterpretation

- Negative sign: high-glue workers in high-AI occupations are displaced, not preserved
- Effect operates through *occupational reallocation*, not within-occupation survival
- Theory-specific prediction $\neq$ generic heterogeneity claim

Claude distinguished theory-specific predictions from generic heterogeneity findings. That distinction determines what the paper can claim.

Division of Labor Across the Full Lifecycle

Task	Human Researcher	AI (Claude)
Core insight	Endogenizes bundle strength via glue	
Literature positioning	Sets the GLW extension	
Variable specs	Co-designed, approved	Elaborated O*NET codes; caught crosswalk inconsistency
Pipeline architecture	Approved the design	Designed 11-script layout from scratch
Claims discipline	Enforces the gate	Drafted allowed/forbidden claims
Adversarial review	Decided which weaknesses were fatal	Structured the 6 adversarial findings
Data execution	Runs the scripts	
Interpretation	Final call on repositioning	
Paper writing	Authors	Assists with exposition

Key Prompting Strategies in Case 4

1. **Specification-driven:** exact O*NET item codes, not vague requests — prevents hallucinated variable definitions
2. **Constraint-based (claims discipline):** allowed and forbidden claims specified before results appear — prevents data mining
3. **Adversarial:** “find 6 structural weaknesses” not “review this memo” — prevents sycophantic validation
4. **Modular decomposition:** each script as an independent testable unit — prevents silent errors accumulating across stages
5. **Deferred execution:** pipeline fully specified before any real data is analyzed — separates design from execution, preserving pre-specification discipline

Each strategy is a structural response to a specific failure mode, not a stylistic preference.

Failure Modes Specific to Case 4

1. **AI agrees too readily:** asking “does this design look good?” produces validation; adversarial mode must be explicit
2. **Vague theory → untestable claims:** “glue work matters” is not a prediction; specification-driven prompting forces operationalization
3. **Data mining disguised as exploration:** claims discipline must be logged before results; retroactive framing is indistinguishable from pre-specification to a reader
4. **Silent crosswalk errors:** the 12% SOC loss is found only because a validation rule checks merge rate; without validation rules, the pipeline produces credible-looking numbers
5. **Mechanical overclaiming:** when data contradicts theory, the temptation is ad hoc reframing; adversarial review forces the repositioning to be deliberate and documented

The Meta-Irony: AI Doing Glue Work

What the paper studies

- Glue work: coordinative labor that binds complex task bundles
- Holds specifications together across stages
- Tracks implicit commitments across time
- Underrewarded because its contribution is diffuse in output attribution

What Claude did in the research itself

- Maintained specification coherence across 11 scripts and 4 research stages
- Tracked which claims were allowed before results arrived
- Structured the adversarial review to produce actionable findings
- Held the interface between theory and empirical design

The AI is demonstrating its own research subject.

That is not an accident — it is evidence about where AI comparative advantage lies in complex intellectual work.

Key Lessons from Case Study 4

- AI can operate across the **full research lifecycle** — theory articulation, pipeline design, adversarial review — not just execution
- The human researcher must own the core insight, literature positioning, data execution, and final interpretation
- **Specification-driven and adversarial prompting** are structural defenses against the most common AI failure modes in research settings
- **Claims discipline** must be established before results appear; it cannot be reconstructed credibly after
- When data contradicts theory, the right move is deliberate repositioning, not ad hoc reframing; AI can structure that repositioning but cannot decide whether it is intellectually honest
- This is a new pattern: **AI as full-lifecycle research collaborator**, distinct from skill, empirical pipeline, or dataset harness

Case Study 5

Inside the sprint — anatomy of an agentic research workflow, April 15–18, 2026

What Case 5 Adds to Case 4

Case 4 showed the product

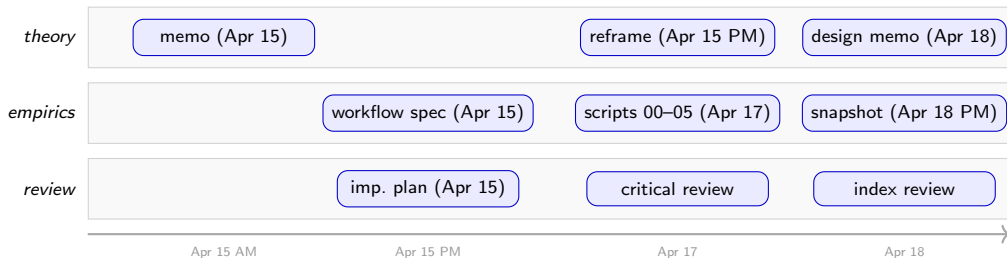
- research question (glue work + AI labor market)
- 2×2 hypothesis
- 11-script pipeline
- adversarial review and repositioning
- division of labor across the lifecycle

Case 5 shows the process

- how the first prompt is written
- how data are chosen and fetched
- how results are read and questioned
- how one index becomes four, then a distribution
- how LLMs themselves generate new data
- how each stage closes with a memo and a critique

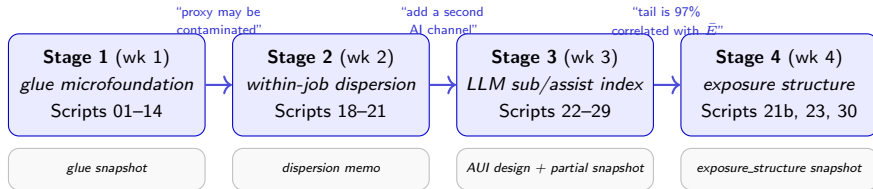
Guiding claim. Case 4 answers “what did the collaboration produce?”. Case 5 answers “how would a student replay it on a different research question?”. **Four stages, eight weeks, one puzzle.** Stage 1 (Apr 15–18) is the four-day first sprint; Stages 2–4 layer on dispersion, an LLM-labeled sub/assist index, and the refined within-job distribution.

The Four-Day Timeline



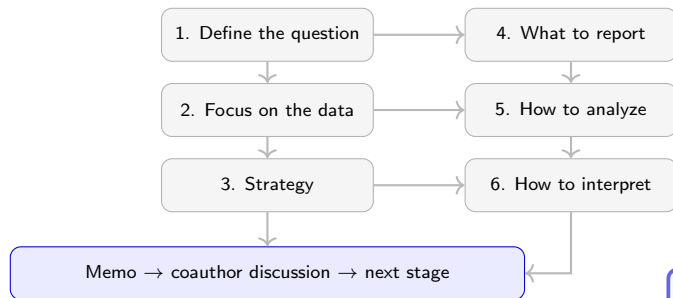
Five acts across three lanes. 1. Framing → 2. Data strategy → 3. Execution → 4. Reflection and refinement → 5. Writing and iteration. Theory, empirics, and review run in parallel, not in sequence. The next 16 frames walk through these five acts inside Stage 1.

The Four-Stage Project Arc



Pattern. Each stage ends with a written memo; each transition is a *coauthor-delivered critique* of that memo. Stages are not planned upfront — each one is scoped from the previous stage's open questions. The workflow is gradual, structured, and human-corrected at every turn.

The Six-Element Stage Template



Every stage reuses the same six elements.

- **Question:** name the specific gap, not the topic.
- **Data:** name the rows and columns, not the dataset.
- **Strategy:** state the identification-*style* claim (descriptive / comparative / causal).
- **Report:** the figures and tables the stage will produce.
- **Analyze:** the spec ladder from simple to preferred.
- **Interpret:** what the result would mean.

Stage 1 (next 16 frames) is the detailed walk-through of this template. Stages 2–4 compress it into 4–8 frames each.

Act 1 — Initiating the Discussion

Opening prompt. Open with the *puzzle*, not the method. The first turn names what is unresolved in the field and invites the agent to push back on the framing before any code or measure is mentioned.

First-turn material, adapted from `glue_work_research_memo.md` §1:

Garicano, Li, and Wu (2026) push back on task-level exposure measures by arguing that labor markets price *jobs*, not isolated tasks, so AI affects occupations through a coordination-cost channel. But their coordination cost is exogenous. This memo argues that **glue work** — the underrewarded coordination labor that preserves shared context across interdependent tasks — is the missing microfoundation for bundle strength.

What the opener buys. The agent now knows: (i) the paper is theoretical, (ii) the key primitive to be endogenized is coordination cost, (iii) the name of the candidate mechanism is glue work. Everything downstream — data choice, proxy design, figure set — is bounded by this frame.

Defining Theoretical vs. Empirical Questions

Theoretical questions

- Can coordination cost c be endogenized as a stock of coordination capital k produced by glue work?
- Does $\lambda < 1$ (partial appropriation of glue's social value) deliver a welfare externality?
- Does a second AI channel a_g acting on coordination tasks weaken bundles that a_1 alone would leave intact?

Empirical questions

- Do occupations with similar AI exposure but different glue content follow different employment trajectories after November 2022?
- Is the divergence stronger for *AI-hard* glue than for coordination in general?
- Are AI exposure and glue content empirically orthogonal enough to identify heterogeneity?

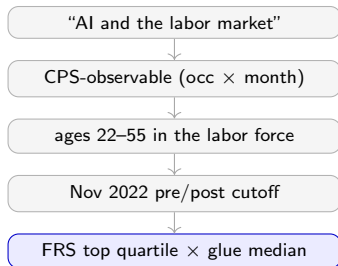
Discipline. The two question sets share vocabulary (“bundle strength”, “coordination”) but not evidence. Theoretical questions are answered by model writing; empirical questions by data. Never let one stand in for the other.

Why Run the Empirics First

- A 62 KB theory memo already existed on Apr 15 21:18. The theory was articulated but not stress-tested.
- Empirical illustration is **cheap to run and expensive to fake**. Running it early exposes any gap between the theory's prediction and what CPS and O*NET actually show.
- Journal readers of a theory paper with no motivating evidence do not take the framework seriously. The motivation has to be visible in the first pages.
- Case 4's headline lesson — “when data contradicts theory, reposition, don't fudge” — is only available *because* the empirics ran early enough to contradict anything.

Ordering rule. Lead with the empirics whenever the theory is already legible. The empirics act as a first round of adversarial review — cheaper than a referee and faster than a coauthor.

Sharpening — From “AI & Jobs” to One Window



Each narrowing is a decision.

- CPS-observable: rules out within-firm reallocation; only occupational movement is visible.
- 22–55: excludes student entry and retirement churn, which move for unrelated reasons.
- Nov 2022: coincident with ChatGPT's public release; visible shock, not a causal cutoff.
- Top quartile × median: keeps Group 4 (high-AI, high-glue) large enough to support heterogeneity.

```
PRE_PERIOD_END      = "2022-10"  
POST_PERIOD_START   = "2022-11"  
AGE_MAIN             = (22, 55)  
MIN_RAW_OBS_PREPOST = 200  
AI_HIGH_QUARTILE     = 4
```

Every narrowing is a line in config.py.

Act 2 — Deciding Which Data

Dataset	Role	Why chosen	Alternative (why not)
O*NET 30.2 Work Activities	glue proxy: 41 task-importance items	task-level, occupation-level, up-dated 2024	O*NET Abilities: less task-specific; cognitive labels drift faster
FRS AIOE (2021)	AI exposure score by occupation	clean SOC 2018 crosswalk; widely cited	Webb (2020): occ1990dd → Census 2018 crosswalk not yet built
IPUMS CPS monthly	employment and entry outcomes	22–55 sample, occupation codes, API access	OEWS: annual only, too aggregated to see monthly shifts
BTOS AI-use Q7	<i>intended</i> Bartik shift-share shock	sector × time AI adoption would identify exposure	Q7 suppressed at sector in all releases except 2 recent biweeks (~17%); rejected

Lesson. Every dataset that makes it into the pipeline has an alternative that was ruled out. Write the rejection reason down — it is the difference between a pipeline and a list of downloads.

Getting Data — API First, Manual Fallback

Automated (preferred)

- `00_download_raw_data.py` fetches O*NET 30.2, the O*NET → SOC crosswalk, the BLS SOC → Census crosswalk, and the FRS AIOE appendix. Idempotent: skips files that already exist.
- `03_build_cps_occ_month_panel.py` uses `ipumspy` and the `IPUMS_API_KEY` env var. It requests an extract, polls every few minutes, and downloads when ready.
- `requirements.txt` pins `ipumspy>=0.4` so the API contract does not drift silently.
- Every fetch writes to `output/logs/download_log.csv`.

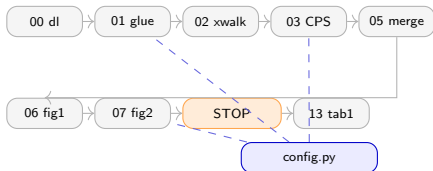
Manual (fallback)

- Webb (2020) AI exposure: MailChimp sign-up on the author's site → emailed Excel file → save to `data_raw/ai_exposure/webb_ai_exposure.xlsx`. Script 07 skips robustness rows if absent.
- IPUMS without API key: browse `cps.ipums.org`, request the same sample, drop the `.dat.gz` + `.xml` into `~/Downloads/Data-Analysis/`. Script 03 autodetects.
- Failed URLs in `download_log.csv` name the destination, not just the URL: a human can resume where automation stopped.

The API/manual split is the one place where the agent should stop and ask the human for credentials. API keys belong in the shell profile, never in a script.

Act 3 — Running the Pipeline from Instructions

The rhythm. 51 unit tests on synthetic data *before* the first download. Each script writes a `.parquet`; later scripts read those, never the raw files. Parameters live only in `config.py`.



STOP GATE before Table 1.

```
=== Figure 2 Inspection ===  
Group 3 (High AI, Low Glue):  
  pre=94.6  post=89.0  d=-5.6  
Group 4 (High AI, High Glue):  
  pre=99.5  post=99.8  d=+0.2
```

- Gate passes: Groups 3 and 4 diverge after Nov 2022; Table 1 is worth computing.
- Gate fails: inspect the proxy first. A regression table before the gate is false precision.

Act 4 — When the Interaction Flips Sign

On Apr 15 22:30 the earliest glue_core_z run produced an uncomfortable pattern:

Glue content **reverses sign** depending on AI exposure:

- Among low-AI occupations: High Glue $\rightarrow$ +27 pp better outcome than Low Glue.
- Among high-AI occupations: High Glue $\rightarrow$ -15 pp worse outcome than Low Glue.

The interaction ($AI \times Glue$) is negative. The simple “glue protects bundles” story predicts a positive interaction.

— critical_review_and_research_directions.md

Two readings

- (a) the theory is wrong;
- (b) the proxy conflates “coordination present” with “coordination dominant” and pools AI-easy documentation with AI-hard care.

Reframe chosen

- Protection operates through occupational *reallocation*, not within-occupation survival.
- Coordination-secondary occupations (analysts) decline; coordination-primary ones (nurses) absorb the reallocation.
- The negative interaction is consistent with the theory once Condition 2 (sufficient coordination capacity) is stated explicitly.

Better Measures — the Four-Index Family

Index	What it captures	Construction	Why it matters
glue_core_z	baseline coordination bundle	5-item coordination average	starting diagnostic, but contaminated
glue_aihard_z	embodied / relational coordination	keeps only human-hard items	drops LLM-easy pseudo-glue
glue_dominance_z	whether coordination is central	glue items relative to full activity bundle	separates primary from secondary coordination
glue_protection_z	replacement-risk composite	weighted mix of hard glue, dominance, social skill, AI-easy penalty	closest summary measure for the paper

Pedagogical point. Stage 1 did not stop at one index. It built a small family of measures so the economist could ask which pattern survives once the naive glue score is decomposed.

Better Measures — AI-Hard vs. AI-Easy Coordination

AI-hard coordination items

- Communicating with supervisors, peers, and subordinates
- Establishing and maintaining relationships
- Assisting and caring for others
- Coaching and developing others
- Resolving conflicts and negotiating

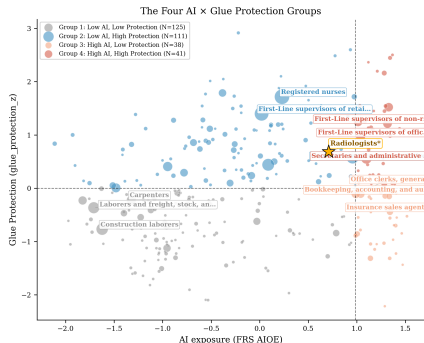
AI-easy pseudo-glue items

- Documenting and recording information
- Updating and using relevant knowledge
- Interpreting information for others

The O*NET codes still live in the project config; the slide keeps the economic distinction rather than the implementation detail.

Why split them? A radiologist and a nurse may both coordinate, but only some coordination is hard for AI to absorb. That is the distinction the refined measures are trying to preserve.

Reading Figures, Suggesting Further Analysis

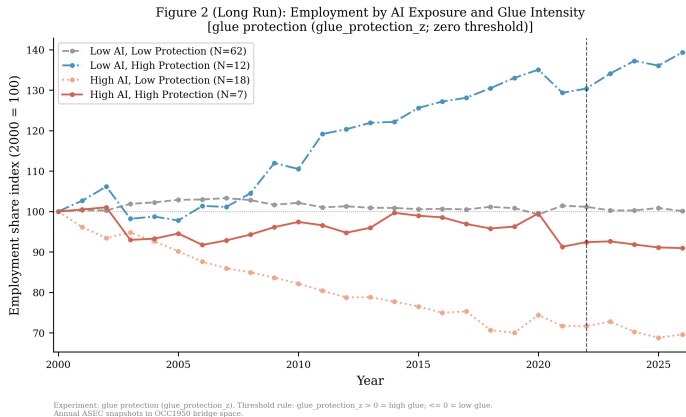


What this quadrant plot triggers.

- AI exposure and Glue Protection are only weakly correlated.
- The high-exposure column already splits into two populations.
- Bottom-right occupations look very different from top-right occupations.
- A single high-AI average would hide the mechanism.

Next move. Compare high-protection and low-protection occupations *within* the high-AI column. That request became the differential figure.

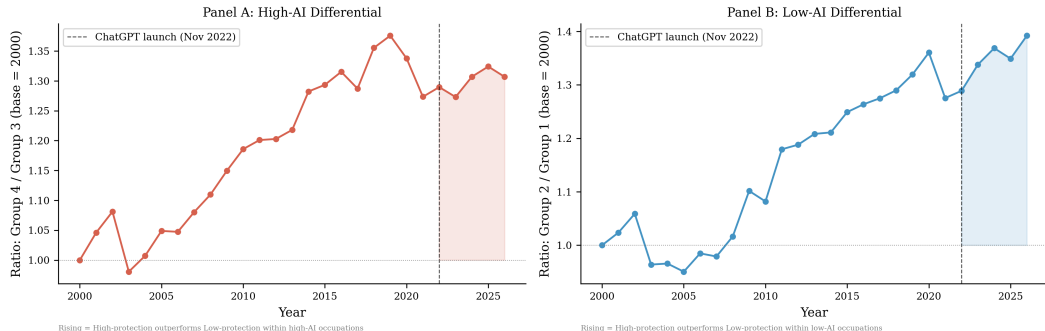
The Headline Figure — Four-Group Employment Trajectories



Reading. Once occupations are split by both AI exposure and Glue Protection, the trajectories are no longer pooled together. The theory claim is descriptive: similar AI exposure can coexist with different long-run employment paths.

The Companion Figure — Inside the High-AI Column

Figure D2: Glue Differential by AI Bucket
(Comparison of high- vs low-protection occupations within each AI bucket)



Groups split using `glue_protection_z > 0` = high glue; `<= 0` = low glue. Employment share ratio normalized to 1.0 in each panel's base year.

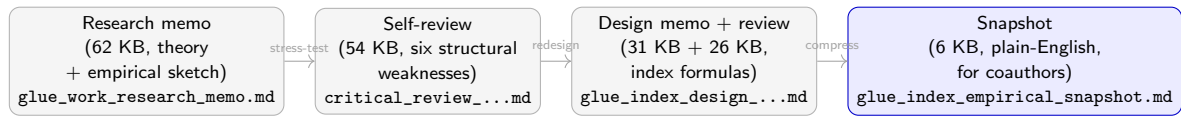
Why this matters. The ratio plot asks the cleaner question: among occupations with high AI exposure, do the high-protection occupations evolve differently from the low-protection occupations? That is the direct diagnostic behind the mechanism story.

The Diagnostic Suite

Diagnostic	What it asks	Where it lives (and what it found)
Management contamination	Is <code>glue_core</code> just measuring whether the SOC prefix is “11”?	Script 01 prints <code>corr(glue, is_mgmt)</code> and warns when > 0.4 .
Sector residual (<code>glue_sector_z</code>)	Is glue just healthcare vs. production?	Partial out major-group means before re-ranking.
O*NET multi-wave (23.3 vs. 30.2)	Did task importance shift differentially in high-AI occupations?	Script 04: $\hat{\beta}_3 \approx 0$, $p \approx 0.52$. Attributed to the 2022–23 O*NET field lag.
Pre-trend decomposition (D1)	Is the divergence post-2022 or part of a long-run drift?	<code>figure_pretrend_allgroups_longrun_annual.png</code> : drift predates 2022.

Rule. Every claim in the snapshot is paired with the diagnostic that could have falsified it. The diagnostic lives on disk with a deterministic name.

Act 5 — Three-Document Writing Architecture



Each step has one job. The memo names the claim. The self-review tries to break it. The design memos rebuild the measure so the claim survives scrutiny. The snapshot delivers the survivor to a coauthor in one page. Writing one document for all four purposes produces a paper that defends everything badly.

Refining — From Memo to Snapshot

Memo opener (argumentative, for the reviewer)

A central question in the economics of AI is whether and how artificial intelligence displaces workers. Garicano, Li, and Wu (2026) push back ...by arguing that labor markets price *jobs*, not isolated tasks. ...This memo argues that **glue work** is the missing microfoundation for bundle strength.

Snapshot bottom line (declarative, for the coauthor)

Two occupations can have the same AI exposure and different labor-market trajectories. The Glue Protection index is an attempt to name one dimension along which they differ. The construction is declarative and inspectable; the next question is how to rationalize the pattern.

- Compression ratio: ~ 62 KB memo $\rightarrow \sim 6$ KB snapshot, roughly $10\times$.
- Audience shift: reviewer (argue for the framework) $\rightarrow$ coauthor (inspect the data).
- Vocabulary shift: “microfoundation” $\rightarrow$ “two occupations”; theoretical verbs $\rightarrow$ empirical verbs.

Spotting Issues, Naming Open Threads

Issue	Where surfaced	Disposition
Bartik shock infeasible	Act 2 data scan (BTOS Q7 suppressed at sector)	deferred; wait for a richer AI-adoption panel
glue_core contaminated by LLM-easy items	Apr 15 self-review	glue_aihard_z promoted to co-primary; glue_aieasy_z added as penalty
Coordination-presence vs. dominance	Apr 17 review	glue_dominance_z added (mean glue $\div$ mean 41 activities)
Webb (2020) not integrated	crosswalk triage (occ1990dd $\rightarrow$ Census 2018 not built)	parked as robustness; waits on a new crosswalk
O*NET multi-wave null ($\hat{\beta}_3 \approx 0$)	Script 04 diagnostic	attributed to 2022–23 field lag; rerun on O*NET 31 when released

Each row is where the next sprint starts. A sprint that ends without a written issue list starts the next sprint by rediscovering its own unsolved problems.

Division of Labor — Move by Move

Move	Agent did	Economist did
Name the puzzle	draft the three-sentence opener from a literature sweep	chose GLW (2026) as the anchor; named glue work as the missing microfoundation
Choose O*NET items	proposed five items from the content model	vetoed three LLM-easy items; added the embodied/relational list
Build the pipeline	wrote scripts 00–14, wired <code>config.py</code> , added 51 unit tests	locked the parameter values and the STOP GATE criterion
Spot sign reversal	printed the pre/post deltas; surfaced the pattern	recognised it as a theory problem, not a bug
Reframe theory	produced three candidate rewordings	chose reallocation-not-survival; wrote the new abstract
Design the four-index family	drafted <code>aihard</code> , <code>dominance</code> , <code>social</code> , <code>aieasy</code> with weights	fixed the 35/25/25/15 weights; named the composite
Write the snapshot	compressed 62 KB → 6 KB on request	read the draft as a coauthor would; rewrote the bottom line

The constant. The agent executes; the economist decides what is true. The economist’s work is not replaced — it is concentrated into judgment calls.

End of Stage 1 — Memo, Critique, and What Triggered Stage 2

The memo (6 KB)

"Two occupations can have the same AI exposure and different labor-market trajectories. Glue Protection names one possible reason. The construction is inspectable; the next step is to explain the pattern."

One page, plain English, readable by a coauthor in three minutes.

The coauthor critique

Apr 18 critique:

- "You're still using the mean of a within-occupation distribution. The *shape* matters."
- "If two tasks have very different exposures, the job gets *unbundled* — not *substituted*."
- "Add a second measure: how dispersed is the exposure *within* an occupation?"

The question that launched Stage 2

Holding mean AI exposure fixed, does within-job dispersion of exposure predict labor-share outcomes?

That question became the Stage 2 workflow over the next week.

Rule. A short memo plus an honest critique is the cheapest way to make the next sprint smarter.

Case Study 5 — Stages 2, 3, 4

From a single glue index to a distribution — and an LLM-built index in between

Stage 2 — AI Task Dispersion (opens)

Question. Given the mean AI exposure $\bar{E}_o$, does within-occupation dispersion of exposure predict labor-share movement?

Data (new). Add **Eloundou et al. (2023)** GPT-4 T -scores as a second, independent AI-exposure source parallel to FRS. Tasks, not abilities. 13,225 Core tasks across 872 O*NET-SOC codes.

Strategy. Per-occupation features (task_count, ai_exposure_sd) built at O*NET-SOC, bridged through SOC 2018 $\rightarrow$ OCC1950 via CPS-weighted pairs. Outcome: Δ share in an OCC1950 $\times$ (age $\times$ education) cell.

Report. Coefficient tables for Experiments A–E across four cells and two sources. Binscatter figures.

Analyze. Nested WLS ladder weighted by pre-period employment share, HC1 SEs:

- M0: $\Delta s \sim \bar{E}$
- M1: $\Delta s \sim \bar{E} + SD$
- M2: $\Delta s \sim \bar{E} + TC$
- M3: $\Delta s \sim \bar{E} + SD + TC$

Interpret. Does $|\hat{\beta}_{SD}|$ or $|\hat{\beta}_{TC}|$ rival $|\hat{\beta}_{\bar{E}}|$ on a 1-SD scale?

Same six-element template as Stage 1; different question.
docs/ai_task_dispersion/ai_task_dispersion_memo.tex, Scripts 18, 18b, 19, 20, 21.

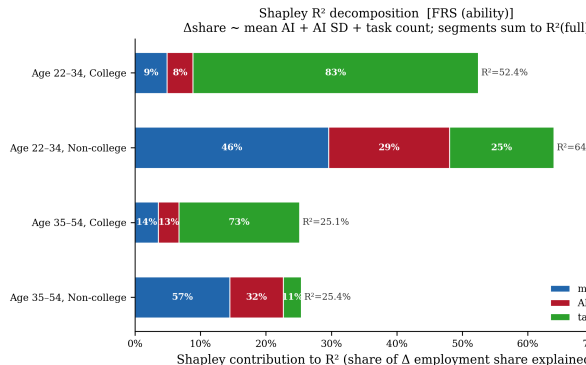
Pipeline home:

Stage 2 — Features, Specs, and the Head-to-Head Question

Feature pairs per source

Source	Features
FRS (abilities)	task_count, ai_exposure_sd — 52 abilities screened at IM ≥ 3
Eloundou (tasks)	task_count_el, ai_exposure_sd_el — Core tasks, T-score 0–4

The head-to-head. All features z-scored within cell so the ratio $|\hat{\beta}_{\text{feat}} / \hat{\beta}_{\bar{E}}|$ says “how many 1-SD of $\bar{E}$ does a 1-SD of the feature deliver?”



Experiment E Shapley decomposition (FRS side). Each cell's full-model R^2 split across $\{\bar{E}, \text{SD}, \text{TC}\}$ by average marginal contribution across orderings.

Stage 2 — Three Findings

1. **Task count is robust.** Young $\times$ College: $\hat{\beta}_{TC} = +0.0059$ ($p = 0.005$) on FRS, $+0.0045$ ($p = 0.002$) on Eloundou. Same sign, both sources, same cell. Carries 62–83% of R^2 in the Shapley decomposition for that cell.
2. **AI SD is cell-specific.** Significant under Eloundou in Young $\times$ College ($p = 0.018$ in M3) and under FRS in Middle $\times$ Non-college ($p = 0.005$). Elsewhere indistinguishable from zero. Not universal.
3. **Mean AI exposure is not a sufficient statistic.** In 5 of 8 cells the Shapley share of the mean-AI channel is below half. The residual share is concentrated in TC or SD depending on the cell.

Middle $\times$ Non-college (FRS; 36.7% of prime-age employment). $\hat{\beta}_{\bar{E}} = -0.0040$ (M1), $\hat{\beta}_{SD} = -0.0041$ (M1). Ratio $|\hat{\beta}_{SD}/\hat{\beta}_{\bar{E}}| \approx 1.01$, and SD adds +15.8pp to R^2 on top of mean AI alone. In this cell, within-job dispersion matches the mean-AI channel dollar-for-dollar.

End of Stage 2 — Memo, Critique, What Launched Stage 3

The memo

ai_task_dispersion_memo.tex
— 14 pages, five experiments,
descriptive throughout.

Bottom line: the task-count finding survives replication; within-job SD is a second channel with cell-specific bite; mean AI exposure is not a sufficient statistic.

The coauthor critique

- “Task count tells you the *number* of tasks. But not whether each task is *AI-substitute* or *AI-assist*.”
- “Eloundou gives one *T*-score per task. You need two: e_{sub} and e_{asst} .”
- “Can the LLM itself do this for 13,000 tasks?”

What launches Stage 3

Build a task-level LLM-labeled index that decomposes AI exposure into substitution and assistance.

This seeds Scripts 22–29 and the unbundling_vulnerability track.

Stage 3 — Unbundling Vulnerability (opens)

The missing decomposition.

Both FRS AIOE and Eloundou T -score deliver *one* number per task or ability. That number bundles two qualitatively different things:

- e_{sub} — AI can *replace* the worker on this task;
- e_{asst} — AI can *help* the worker do this task faster.

A scholar with one task at $e = 0.8$ and a call-center agent with one task at $e = 0.8$ have the same AIOE score — and likely very different labor-market trajectories.

What the LLM can do that the literature can't.

The O*NET task description plus a rubric is enough for a modern LLM to distinguish the two readings task-by-task. 13,225 tasks $\times$ (sub, asst) scores is $\sim 26,000$ new data points that do not exist in any public release.

Deliverables.

- $R_o = E_{\text{sub}} / (E_{\text{sub}} + E_{\text{asst}} + \epsilon)$ — occupation-level substitution share.
- U_o — composite Unbundling Vulnerability Index that uses R_o .

The LLM can score something the original index was never built to see. Home: [docs/unbundling_vulnerability/](https://docs.unbundling_vulnerability/).

Stage 3 — The Ambition: Scoring 13,225 Tasks

Corpus size	13,225 O*NET Core tasks
Per-task output	$(e_{\text{sub}}, e_{\text{asst}}, \text{rationale})$ in $[0, 1]$
Calls required (naïve)	13,225 LLM calls
Budget at Claude Max	one 5-hour window
Strategy	bulk + adjudicate + cache

Three design moves that turn a 13k-call ambition into a tractable pipeline:

- **Bulk:** 10 tasks per Haiku call — one JSON array in, one JSON array out. Cuts calls by $10\times$.
- **Adjudicate:** Sonnet reruns the tasks Haiku flagged as borderline.
- **Cache:** one file per task on disk; re-runs skip already-labeled tasks. Interruption-proof.

Why each move matters

Without bulking: $13\text{k calls} \times \sim 10\text{s} = \sim 36 \text{ hours}$ — no single Claude Max window fits.

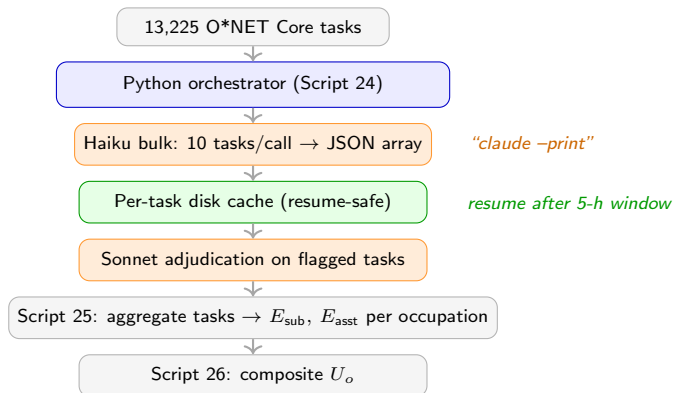
Without adjudication: Haiku's borderline judgments carry the whole index; no second opinion, no way to flag hard cases.

Without caching: any network blip, quota limit, or power loss destroys hours of work. The 5-hour window *will* expire mid-run — the cache turns that from a catastrophe into a coffee break.

Script home.

`scripts/24_label_substitution_assistance.py`

Stage 3 — The Labeling Architecture



Three layers of LLM. Claude Code writes Python (layer 1); the Python spawns `claude -print` Haiku sessions (layer 2); Sonnet adjudicates the borderline cases (layer 3). **11,171 of 13,225 tasks labeled (84.5%)** in one 5-hour window before the quota expired. Resume via cache; no task re-labeled.

Stage 3 — The Rubric Behind the Labels

Replacement score e_{sub}

- 0.00: AI cannot replace the task
- 0.25: AI handles only narrow pieces
- 0.50: human and AI split the task
- 0.75: AI does most of the work; human reviews
- 1.00: AI replaces the task almost entirely

Assistive score e_{asst}

- 0.00: no productivity gain
- 0.25: speeds up a narrow sub-step
- 0.50: roughly doubles throughput
- 0.75: handles most of the time cost
- 1.00: task becomes trivial once AI is used

Three anchors. Clear substitute: *transcribe audio*. Clear assistant: *perform laparoscopic surgery*. Borderline case: *draft a legal contract clause*. Input was a batch of 10 tasks; output was structured JSON with scores, rationale, and an ambiguity flag.

Stage 3 — Token Engineering at Scale

Operational trick

Run `claude -print` from a clean temporary directory so the project's 12k-token `CLAUDE.md` does *not* auto-load into every bulk call.

Why it matters

- 13k tasks means thousands of repeated API calls.
- Small prompt waste becomes massive token waste.
- Cheap calls stay cheap only if the context stays minimal.

Economic interpretation

Saved scale. About 160M tokens avoided over the full run. At this scale, prompt engineering is not cosmetic; it is cost control and latency control.

Teaching point

- The rubric stayed constant.
- The execution environment changed.
- Agentic AI is strongest when the human specifies the rule once and the system executes it thousands of times.

Stage 3 — Running Massive Work Unsupervised

What the economist did.

- Wrote the rubric (~30 lines) and three calibration anchors.
- Read ~12 LLM-returned labels by eye to sanity-check the scale.
- Launched the run and walked away.

What Claude Code did.

- Wrote the orchestrator (Script 24).
- Spawned thousands of `claude -print Haiku` sessions.
- Wrote each task's result to its own cache file.
- Detected quota expiration, logged cleanly, exited gracefully.

What the LLM (Haiku/Sonnet) did.

- Returned $(e_{\text{sub}}, e_{\text{asst}})$ for 11,171 tasks on the first pass.
- Flagged 3,456 ambiguous tasks for adjudication.

Log excerpt (`script24_bulk_run.log`):

```
06:41:22 INFO  batch 1108 (10 tasks) -> 10 cached
06:41:41 INFO  batch 1109 (10 tasks) -> 10 cached
06:42:01 INFO  batch 1110 (10 tasks) -> 10 cached
...
11:28:03 INFO  rate-limit backoff (45s)
11:28:48 INFO  batch 1117 (10 tasks) -> 10 cached
11:29:57 ERROR claude-max window expired
11:29:57 INFO  cache now holds 11,171/13,225 tasks
11:29:57 INFO  resume via: python scripts/24_...py --
                    skip-gates
```

Supervision cost. ~ 20 minutes of economist time to spec the rubric. ~ 5 hours of unsupervised LLM work. ~ 12 labels read by eye out of 11,171. This is the ratio that makes LLM labeling a viable empirical tool at corpus scale.

Stage 3 — From per-Task Scores to U_o

Task → **occupation aggregation** (Script 25):

$$E_o^{\text{sub}} = \sum_t w_t \cdot e_t^{\text{sub}}, \quad E_o^{\text{asst}} = \sum_t w_t \cdot e_t^{\text{asst}}$$

$$R_o = \frac{E_o^{\text{sub}}}{E_o^{\text{sub}} + E_o^{\text{asst}} + \epsilon}$$

Composite (Script 26):

$$U_o = \alpha \bar{E}_o + \beta V_o + \theta T_o + \lambda (R_o \cdot \bar{E}_o) - \gamma C_o - \mu Q_o$$

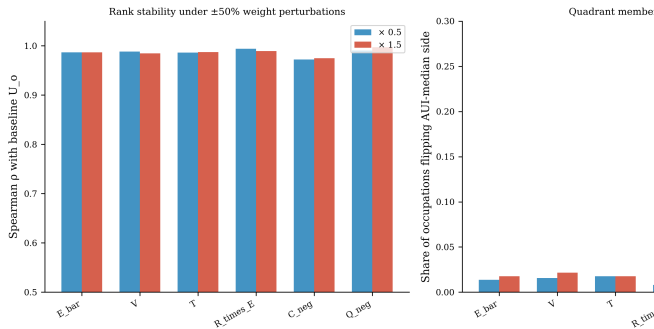
Interpretation. V is within-job dispersion, T is top-tail exposure, C is coordination cost, and Q is accountability intensity. The point is not that one scalar is perfect; it is that the index records several reasons a job may be vulnerable or protected.

Stage 3 — Weight Sensitivity of U_o

Declared weights

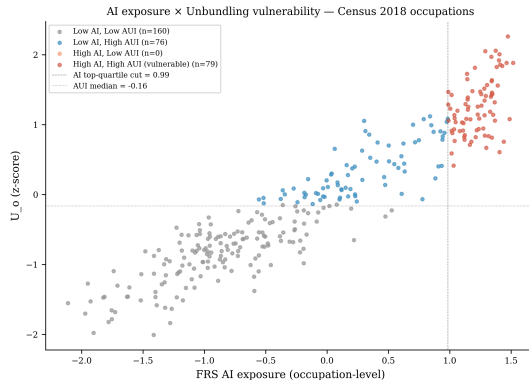
(config.py:AUI_WEIGHTS, not data-fit):

```
AUI_WEIGHTS = {  
    "E_bar": 0.30,  
    "V": 0.15,  
    "T": 0.20,  
    "R_times_E": 0.15,  
    "C_neg": 0.15,  
    "Q_neg": 0.05,  
}
```



Stability check. The top and bottom occupations in U_o survive reasonable re-weighting. That does not prove the weights are correct; it shows the ranking is not driven by one arbitrary coefficient choice.

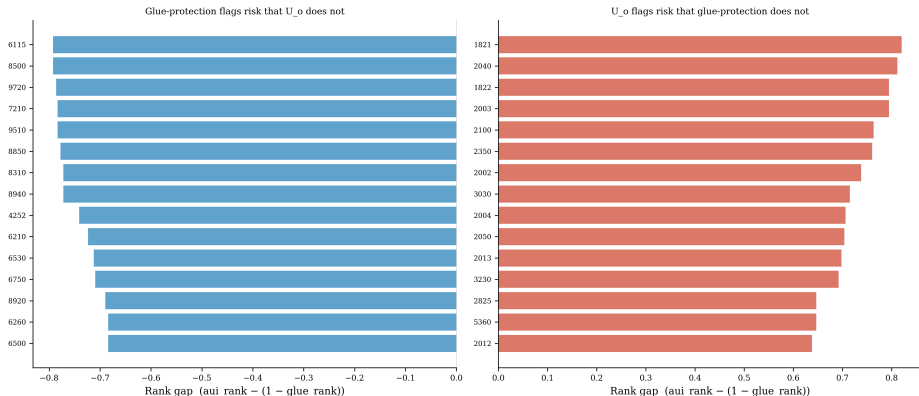
Stage 3 — Reading the U_o Index



Quadrants. Top-right occupations are wholesale-substitution candidates. Top-left occupations face reskilling pressure even when mean exposure is not especially high. The decomposition changes the stories at the margin, not only at the extremes.

Stage 3 — Where U_o Changes the Ranking

Where P^{rich} and U_o disagree



Disagreement region. The interesting cases are where U_o and P^{rich} disagree. That is exactly where the task-level labeling work pays off: it adds information the Stage 1 protection index could not recover.

End of Stage 3 — Memo, Critique, and Cost

The memo (partial)

unbundling_vulnerability_empirical_s
is populated *partially* because
Stage 3 paused with 2,054
remainder + 3,456 adjudication
queued.

The design memo + LLM prompts
+ quadrant figures are complete;
 R_o numbers are placeholder pend-
ing the remaining labels.

The coauthor critique

Apr 22 call, Xiaojie Liu:

*"Your top-tail share is 0.97 corre-
lated with mean exposure. You
still do not have an independent
axis. The within-job shape of the
distribution, not the upper tail, is
where you can learn something $\bar{E}$
does not carry."*

What launches Stage 4

*Refine within-job distribu-
tion features; test each one
against mean AI exposure.*

Cost accounting.

~11k tasks $\times$ ~2 Haiku calls
 $\times$ ~300 tokens $\approx$ 7M tokens in
one 5-h window. Cache guaran-
tees no re-runs.

Stage 4 — Exposure Structure Refinement (opens)

Question. Holding mean AI exposure $\bar{E}$ fixed, which *distribution-shape* statistics of within-job exposure add information for labor-share outcomes?

Data (no new source). Same FRS + Eloundou inputs. What changes is the feature set: instead of just mean and SD, compute a family of shape statistics.

Strategy.

- Tighten tail thresholds: $p_{33} \rightarrow p_{10}$ (FRS), $T \geq 3 \rightarrow T = 4$ (Eloundou).
- Add `top_bottom_gap`, `top_within_mean`, `bottom_quantile_mean`, `importance_rank_slope`.
- Regress Δshare on each feature (conditioning on $\bar{E}$), pooled and per-cell.

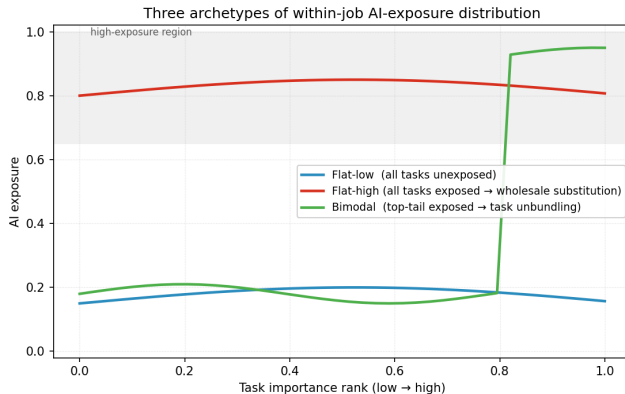
Report.

- Correlation-with- $\bar{E}$ table for every feature.
- F1 pairwise ΔR^2 rankings.
- F5 focused horse race: $\beta(X)/\beta(\bar{E})$ ratios.
- Shapley-on-4 allocating the increment over the $\bar{E}$ -only baseline.

Analyze. Cluster-robust SEs at occ1950 for pooled specs (each occupation contributes four cell-rows; HC1 understates). Script `21b_run_exposure_structure_regressions.py`.

Interpret. Which refined features beat $\bar{E}$ in magnitude? Which disappear once $\bar{E}$ is controlled?

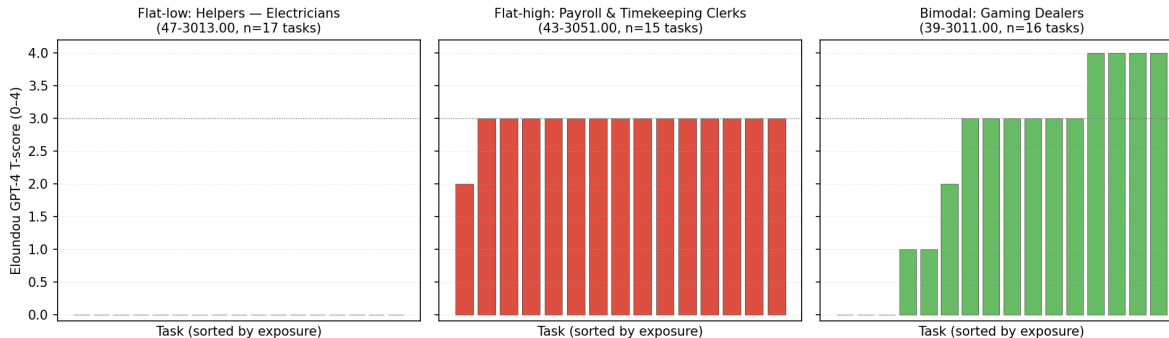
Stage 4 — Three Archetypes of Within-Job Exposure



Conceptual point. Flat-low, flat-high, and bimodal jobs can share the same mean exposure while implying three different labor-market predictions: untouched, wholesale substitution, and unbundling.

Stage 4 — Real Analogues for the Three Archetypes

Real-occupation analogues of the three archetypes (Eloundou tasks)



Examples. Helpers–Electricians look flat-low, Payroll and Timekeeping Clerks look flat-high, and Gaming Dealers look bimodal. The point is to map an abstract distribution shape back to recognizable occupations.

Stage 4 — The Refined Feature Set

Feature	What it measures	r^2 on $\bar{E}$ (FRS)
top_bottom_gap	top-decile mean — bottom-decile mean	0.001 (nearly orthogonal to $\bar{E}$)
glue_protection_z	P^{rich} from Stage 1	0.040
ai_exposure_cv	SD / mean	0.046
top_tail_mean_frs_p33	mean exposure above 67th pct	0.049
task_count	number of relevant tasks	0.072
top_tail_share_frs_p10	weighted share above 90th pct (tightened)	0.882
top_tail_share_frs_p33	weighted share above 67th pct (legacy)	0.933

```
# config.py, added 2026-04-22
TOP_TAIL_PERCENTILE_FRS_STRICT    = 0.10    # upper-decile (strict)
TOP_TAIL_PERCENTILE_FRS_LEGACY    = 1/3      # upper-tercile (legacy)
TOP_TAIL_THRESHOLD_ELOUNDOU_STRICT = 4       # T-score == 4 (strict)
BOTTOM_QUANTILE                   = 0.10      # symmetric bottom-decile
```

Stage 4 — F5 Horse Race vs. the Literature's $\bar{E}$

Pooled M4: $\Delta\text{share} \sim \bar{E} + 4 \text{ features} + \text{cell FE}$. Cluster-robust SE at occ1950. All z-scored.

Source	Variable	$\hat{\beta}$	p	$ \hat{\beta}/\hat{\beta}_{\bar{E}} $	Note
FRS	$\bar{E}$ (literature index)	-0.00125	0.041	1.00	baseline
FRS	task count	+0.00268	0.001	2.13×	<i>opposite sign, 2× bigger</i>
FRS	AI CV	-0.00183	0.038	1.46×	same sign
FRS	top-bottom gap	+0.00052	0.505	0.41×	not sig.
FRS	glue protection	-0.00011	0.826	0.09×	absorbed by $\bar{E}$
Eloundou	$\bar{E}$ (literature index)	-0.00325	0.019	1.00	baseline
Eloundou	AI CV	-0.00296	0.021	0.91×	<i>same sign, matches $\bar{E}$</i>
Eloundou	task count	+0.00152	0.069	0.47×	borderline
Eloundou	top-bottom gap	-0.00088	0.223	0.27×	not sig.
Eloundou	glue protection	-0.00042	0.487	0.13×	absorbed by $\bar{E}$

Reading. With every feature on a 1-SD scale, $|\hat{\beta}_{\text{feat}}/\hat{\beta}_{\bar{E}}|$ says “how many 1-SD of $\bar{E}$ does a 1-SD of the feature deliver?” **Task count beats $\bar{E}$ by 2× on FRS (opposite sign); AI CV is 1-for-1 with $\bar{E}$ on Eloundou.** Mean AI exposure is significant, but not the largest channel.

Stage 4 — Shapley-on-4: Where the Extra Explanatory Power Sits

Shares of the $M_4 - M_0$ gap in R^2

	task count	CV	gap	glue
FRS	92%	5%	2%	1%
Eloundou	39%	38%	21%	2%

FRS. Task count is essentially the whole story. The other three features are absorbed into $\bar{E}$.

Eloundou. A two-channel story: task count and CV share the top almost equally, with the gap adding a real third contribution. Glue protection is near zero on both sides.

Glue protection washes out at OCC1950 pooled — but it performed well at Census 2018 in the main cross-section pipeline (Stage 1). **Which axis an index reads correctly is scale-dependent.** The honest reading: glue protection is a Census-2018 object, not a long-run OCC1950 object.

What the regression bought.

- Joint F on the four features: $F(4) = 5.41$, $p = 0.0006$ on FRS; $F(4) = 6.76$, $p = 0.0001$ on Eloundou. The four together clearly add information beyond $\bar{E}$.
- The Shapley decomposition is how we know which feature is doing the work.

End of Stage 4 — Memo and Open Threads

The memo.

docs/exposure_structure/exposure_structure_empirical_snapshot.md

— variable definitions (self-contained), feature correlations with $\bar{E}$, $V \times \bar{E}$ non-linearity, F1–F5 regression block, ratio tables.

What survived.

- `task_count` (FRS) and `ai_exposure_cv` (Eloundou) add information beyond $\bar{E}$ at the pooled aggregation.
- `top_bottom_gap` and `glue_protection_z` wash out at OCC1950 but may still matter cell-specifically or at Census 2018.

Open threads (explicit).

empirical_snapshot.md 16% short. Resume from cache to finish 2,054 + 3,456 labels.

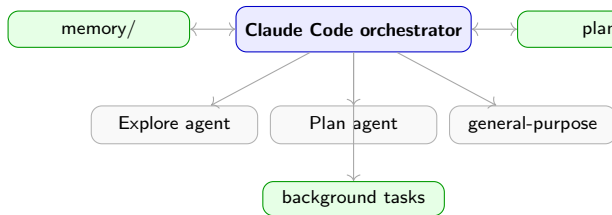
- Labor-market outcome regressions not yet re-run with the completed U_o .
- No causal design — still descriptive motivation for the theory paper.
- Composite AUI weights are declared, not fit. A data-driven re-weighting (future stage) is an option the snapshot flags but does not take.

Each row is where the next sprint starts. The list of open threads *is* the stage's deliverable, just as much as the snapshot itself.

Case Study 5 — Meta-Commentary

Agentic AI under the hood, and three ways to use an LLM

Agentic AI Under the Hood



What ran autonomously during the 8 weeks.

- **Plan mode.** Each stage drafted, edited, user-approved, executed. Plans live at `~/ .claude/plans/*.md` across sessions.
- **Sub-agents.** Explore agent spawned in parallel (sometimes 3 at once) to map the codebase before designing new scripts.
- **Memory.** Per-project files (`project_*.md`, `MEMORY.md`) persisted what was done and where the track paused.
- **Background tasks.** Scripts 19 and 21b runs used `run_in_background + Monitor`, so planning continued while regressions compiled.

None of these were explicitly requested turn-by-turn. Claude Code *chose* when to launch a sub-agent, when to save a memory, when to start a background task. The economist's job shifted from "drive each step" to "approve each plan, critique each memo."

Where Reusable Skills Would Tighten the Loop

Proposed skill	What it would do	Where it saves time
index-builder	Emit a new Script 2X with caching, z-scoring, and claims-discipline notes baked in.	Stage 1, 2, and 4 script scaffolding.
regression-runner	Given an outcome, features, and cells, emit the full F1–F5 regression ladder.	Stage 2 and Stage 4 regression boilerplate.
label-corpus	Wrap the bulk + adjudicate + cache pattern for large task-labeling jobs.	Turns similar corpus tasks into 1-day projects instead of 1-week projects.

Interpretation. The biggest gains now come from packaging repeated research patterns, not from asking the model to be smarter in a single conversation.

Where Reviewer Agents Would Tighten the Loop

Proposed agent	What it would do	Where it saves time
memo-writer	Enforce the six-element template on every end-of-stage memo and auto-link prior outputs.	Stage-close memos across the whole project.
coauthor-critic	Read the memo and surface the top missing checks, as a simulated reviewer.	Replaces part of the coauthor burden; never all of it.

Claim. Claude Code was already sufficient without custom tooling. The next 10× gain would come from making these repeated reviewer patterns reusable.

The Power of LLMs in this Project — Three Uses

1. LLM as pair programmer

Writes Python, runs it, reads the output, iterates on user prompts. The entire pipeline (Scripts 01–30, 21b) falls here. Familiar from Cases 1–4.

Where it fires: every new script, every diagnostic, every figure.

2. LLM as reviewer

Critiques the empirical snapshot, proposes the next design. Stage 1's self-review memo (54 KB); the Apr 22 distillation of the coauthor critique into the Stage 4 question.

Where it fires: every end-of-stage frame in this deck.

3. LLM as empirical data producer

Stage 3: per-task (e_sub, e_asst) labeling. 11,171 / 13,225 tasks labeled autonomously.

An LLM-labeled dataset of 10^4 observations is a first-class empirical artifact — versionable, cross-validatable, publishable.

The third use is the one most often missed. A student who has only seen LLMs as writing assistants has not yet seen LLMs doing labor that was previously the province of thousands of MTurk workers or hundreds of RA-hours. That is the step change this case study is built around.

Ten Lessons from Case Study 5

From Stage 1 (first sprint):

1. **Lead with the puzzle, not the method.** First turn names what is unresolved; the method is downstream.
2. **Empirics first buys theoretical honesty.** The data are the cheapest critic the theory will ever meet.
3. `config.py` + **STOP GATES** = audit-grade pipeline: parameters on disk, inspections at named checkpoints.
4. **When the interaction flips sign, audit the proxy before revising the theory.** A bad measure can imitate a broken mechanism.
5. **A one-page snapshot** costs less than one rerun and compounds more than any single figure.

From Stages 2–4 (this extension):

6. **The first sprint is never the last.** End-of-stage memos + honest coauthor critiques are the lowest-cost lever for the next sprint.
7. **Orthogonality to $\bar{E} \neq$ predictive power.** `top_bottom_gap` has r^2 on $\bar{E}$ of 0.001 but fires only in one cell. `task_count` is correlated with $\bar{E}$ but wins the FRS horse race.
8. **Joint F-tests understate power under multicollinearity.** Shapley R^2 decomposition separates collinear signal from noise.
9. **LLMs are economically viable corpus labelers.** Bulk (Haiku) + adjudication (Sonnet) + per-task cache = 11k tasks labeled in one 5-h window.
10. **Index design is scale-dependent.** `glue_protection_z` survives at Census 2018 but washes out at OCC1950 pooled. Declare the aggregation up-front.

Transferable pattern. Replace “glue work + AI labor market” with any puzzle that has (i) a theoretical primitive, (ii) a public dataset, (iii) a visible post-event window, and (iv) a task corpus an LLM can score. The same four-stage scaffold runs on any of them.

Synthesis and Exercises

Comparing reusable skills, empirical pipelines, public-web datasets, and full-lifecycle collaboration

Compare the Five Cases

	Case 1	Case 2	Case 3	Case 4	Case 5
Main artifact	review report	panel + figures	metadata dataset	repositioned paper	workflow walk-through
Workflow type	reusable skill	3-prompt session	batched public-web	full lifecycle collab.	meta / process audit
Main validation	quote accuracy	ext. benchmark	provenance	claims discipline	replayability on a new question
Main failure mode	hallucinated criticism	wrong crosswalk	weak sources	data-mining / sycophancy	pattern-matching to Case 4 instead of learning its method

When to Use Which Workflow

Build a skill

- repeated use
- high QC needs
- stable output format

Stay ad hoc

- one-off empirical job
- fast iteration
- small prompt set

Build a dataset harness

- many entities
- uneven sources
- provenance matters

Full-lifecycle collab.

- novel research question
- theory needs operationalization
- claims discipline required
- adversarial review needed

And Case 5? Case 5 is not a fifth workflow to pick for a project. It is a meta-case: a walkthrough of Case 4's process so students can replay the same five-act scaffold on their own question. Pick Cases 1–4 to work; read Case 5 to learn how.

The Division of Labor Across All Three

Economist	Agent
define the question and success criteria	execute bounded tasks repeatedly
set the coding and validation rules	manipulate files, prompts, and reruns
spot implausible outputs	surface drafts, diagnostics, and plots
interpret the economic meaning	accelerate the path from question to artifact

The constant: the question, the benchmark, and the interpretation remain human work.

Failure Modes to Watch For

1. **Case 1:** confident but unsupported criticism
2. **Case 2:** clean-looking code with a wrong variable mapping
3. **Case 3:** a rich-looking dataset built on weak or forbidden sources
4. **Case 4:** AI validates the design instead of challenging it, or data contradicts theory without adversarial structure to surface it
5. **All cases:** hidden assumptions that never made it into project files

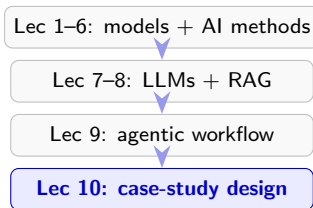
The cure is the same in all four cases: name the validation step, save the rules on disk, and inspect the output like a referee.

Exercises and Deliverables

Exercise	Artifact	Deliverable
Exercise A: MEPS lab	harmonized panel and validated figures	in-class guided run with the existing notebook
Exercise B: fellows assignment	batch-level metadata dataset with provenance	normalized CSV, four figures, and a memo

- Exercise A teaches empirical execution and validation
- Exercise B teaches controlled public-web data construction
- Together they cover two distinct patterns students will actually face in research

Looking Back at the Course



Final message: the distance from question to artifact has collapsed, but economists still own the standards for what counts as evidence.

Appendix

Operational details moved out of the main narrative

Appendix A: Paper-Review Skill Internals

Component	Role
SKILL.md	orchestrates the 7-phase pipeline
shared-directives.md	confidence gate, tone, quote rules
section-prompts.md	specialized prompts by section type
editorial-prompt.md	final quality gate before output

Reason this is appendix material: students need to understand the workflow design before they need a file inventory.

Appendix B: MEPS Prompt Skeleton

Prompt A: download + parse + harmonize + save panel

Prompt B: compute weighted rates by year and subgroup

Prompt C: generate annotated publication-quality figures

Operational note: the notebook remains the handout for the in-class lab.

Appendix C: Fellows Normalized Schema

Column group	Examples
seed fields	name, affiliation, election year
source fields	official profile URL, bio URL, CV URL
biography fields	birth date raw, birth year, PhD year, gender explicit
research fields	research fields raw, normalized field bucket
audit fields	source_birth, source_phd, source_gender, source_field, notes

Appendix D: Fellows Batching Rule

- official roster parsed into 18 batches of 50
- students complete one batch at a time
- every batch must report coverage before interpretation
- if birth-year coverage stays low, use years since PhD instead of age

Appendix E: Resources

- Lecture 10 MEPS notebook in this folder
- Exercise10_ES_Fellows/ script bundle in this folder
- Official Econometric Society current fellows page, checked April 13, 2026
- MEPS public-use files and external benchmark sources used in Case Study 2